Assignment 1
Building a Firewall

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This document describes the network layout for the IA 462 lab virtual network environment and the installation process for PFsense Firewall. Included in this document are the subnet details for the network, and the process I followed to install the firewall. I will also review issues I encountered while building the lab, and the troubleshooting steps I took to resolve them.

Network Design

Virtual Environment

The virtualization software I chose for this lab environment is VMware Workstation. I've worked with both VMware and VirtualBox in the past but went with VMware as it's used in the organization I work for. The software version I'm using is VMware Workstation Pro 17.0.1.

The network consists of two subnets, one for Windows-based operating systems and one for Linux-based operating systems. In Virtual Network Editor, I renamed the host-only networks to Windows and Linux. I unchecked the option "Use local DHCP service" as the firewall will provide DHCP services. I renamed the network adaptor to Windows and Linux. In properties, I went to the IPv4 settings to change the virtual network adaptor's IP addresses.

The network has been configured as follows:

Windows Subnet

- Host IP Adaptor – 192.168.156.120
- Network Subnet – 192.168.156.0/25
- Gateway/Broadcast IP – 192.168.156.1
- Broadcast IP – 192.168.156.127

Linux Subnet

- Host IP Adaptor – 192.168.156.140
- Network Subnet – 192.168.156.128/25
- Gateway – 192.168.156.129
- Broadcast IP – 192.168.156.255

Firewall Installation

For the firewall, I decided to install PFsense. I began the installation by creating a new virtual machine in VMware. I named the virtual machine IA455pfsense. I created the VM with 1 processor, 1024MB of RAM, 20GB storage, and three network cards. I used an older ISO version of PFsense that I had on my computer, then upgraded it once installed to 2.6.0.

The first network connection is configured for NAT. I added two additional network cards for the Windows and Linux subnets.

After I configured these settings, I powered on the system. The first piece of information I entered was the interface name for the WAN connection, em0. After that I was able to configure the settings on the firewall.

Firewall Configuration

I configured the IP address on the LAN interface using option two. I named the interface em1, following the NAT interface, em0. I set a static IP of 192.168.156.1 for the LAN interface, using option 2 from the menu. After adding the IP address, I chose the subnet mask of 25. I did not add an upstream gateway as this LAN interface will be the gateway for the network. I also did not assign an IPv6 address. I chose not to enable DHCP through the firewall.

In order to use the first address of the subnet range for PFsense, I first had to change the IP address of the Windows network host adaptor, which was automatically assigned to this address. I changed this IP address to 192.168.156.120.

Once this step was completed, I connected to the web interface via http://192.168.156.1/. Once I logged into the web interface, it had me complete setup. I named the firewall PFsense, and I added the domain name ia455.advossec.com. I set up a password for the admin and installed updates.

In the interface settings, I named the first LAN connection Windows. The second LAN was configured as Linux with an IP address of 192.168156.129 and a subnet of 25. I added a rule to the Linux firewall to allow network traffic. I went to Package Manager and installed Open-VM-Tools.

Troubleshooting

Installing PFsense

I was unable to install PFsense version 2.6.0 that was downloaded from the internet. The system would reboot after install but then couldn’t find an operating system. I was, however, able to install an older version, 2.4.5 without any issues. Once the system was installed, I updated it to the recent version.

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After installing I learned this was due to a bug in the FreeBSD 10.1 release, where FreeBSD can't boot due to the UEFI bootvar entry not existing in NVRAM. The operating system should install if booted from legacy BIOS, not UEFI. This can be checked while creating a new virtual machine, or by changing the Firmware Type in the VM settings.

Setup Errors

I was unable to log into the web interface after setting up PFsense. I was able to ping the local machine from the PFsense terminal. When initially setting up the network, I tried to configure my WAN connection to be separate from the one my other lab was using. I wasn't sure if two separate labs using the same NAT connection would be able to communicate with each other, which I didn't want. I could not install two NAT networks on the virtual editor. So, I set the WAN connection to NAT as well (the same one my other lab was using). I also changed my other lab setup to have the same subnet mask. After this I was able to login to the PFsense web interface.

Possible Install Issues

Boot from Hard Drive

An issue that may arise during the installation of the firewall is that it does not keep your setup configurations and restarts the installation process. You would want to go to VM settings and change the CD/IDE to Use Physical Drive rather than the ISO image.

Booting from USB

If you decide to create installation media for installing PFsense you may encounter issues. Certain combinations of USB sticks and ports may not work together. You would want to be sure you're using a 2.0 memstick in a 2.0 port.

Remove Unnecessary Hardware

To save resources consider removing unnecessary hardware from virtual machines such as the printer and audio. This will help improve performance.